

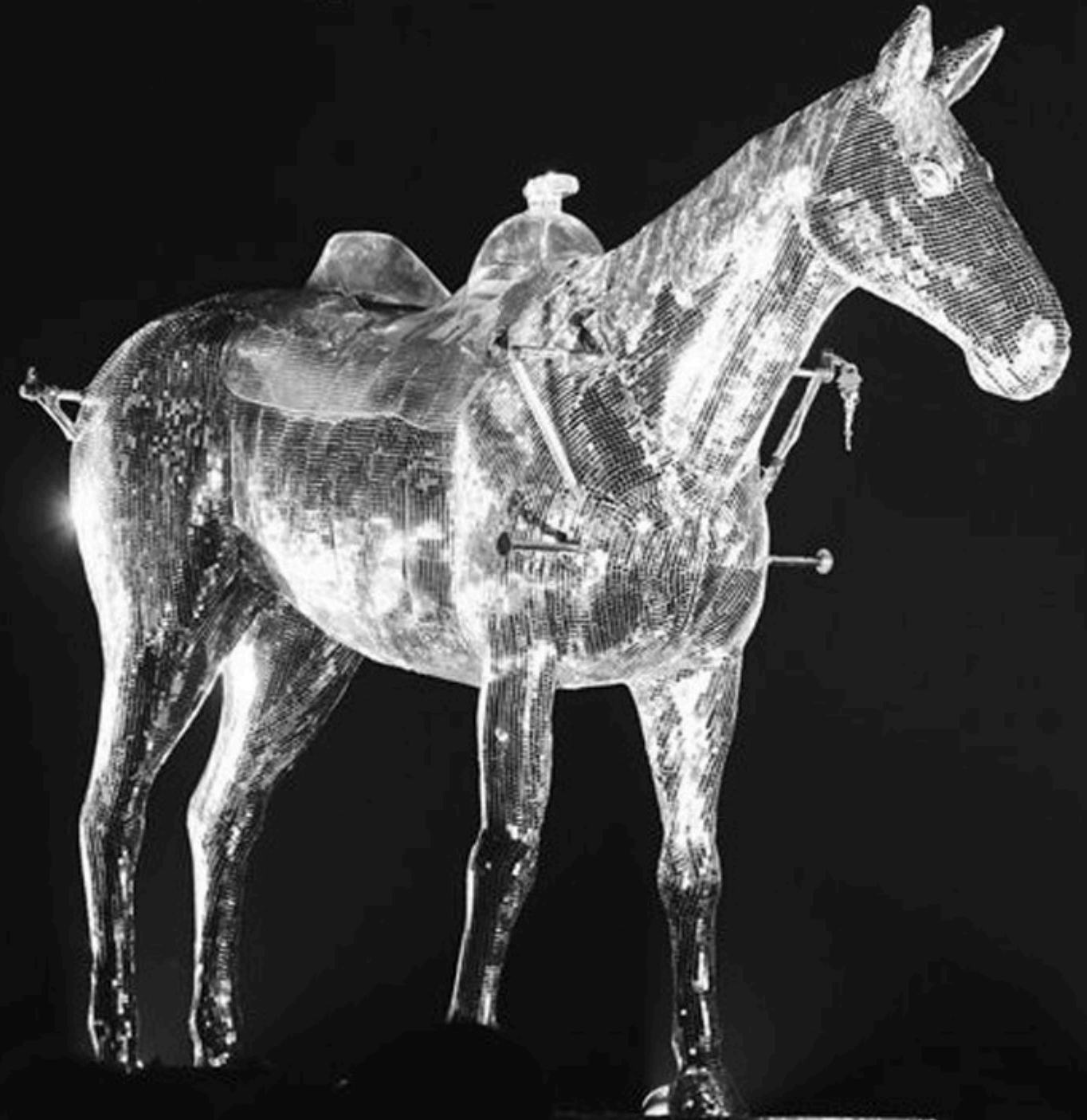
‘YOU WON’T BREAK MY SOUL’ :

*[OR MY SAVINGS ACCOUNT]

THE BEYCONOMICS OF TICKET BUYING

A DATA-DRIVEN GUIDE TO
NAVIGATING THE SECONDARY MARKET

PRESENTED BY
MERCY EME



[1] INTRODUCTION

BRIEF OVERVIEW OF
CASE STUDY

[2] APPROACH

HOW DATA WAS
COLLECTED AND
METHODS USED

[3] RESULTS

INCLUDES PLOTS,
GRAPHS, AND
RELEVANT CODE

[4] DISCUSSION

IN DEPTH ANALYSIS+
RECOMMENDATIONS

[5] LIMITATIONS

SETBACKS+POTENTIAL
IMPROVEMENTS

[6] CONCLUSION

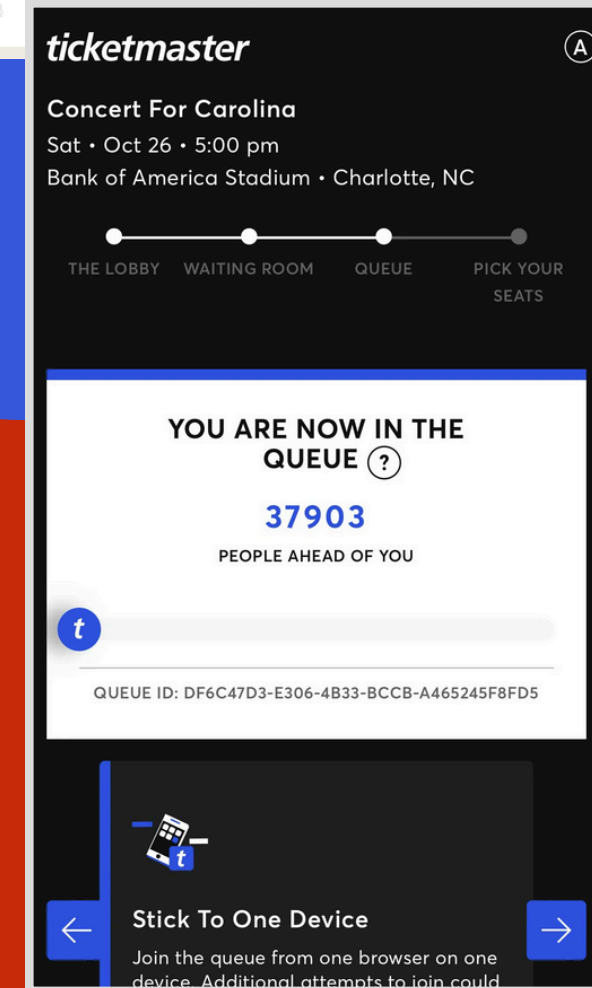
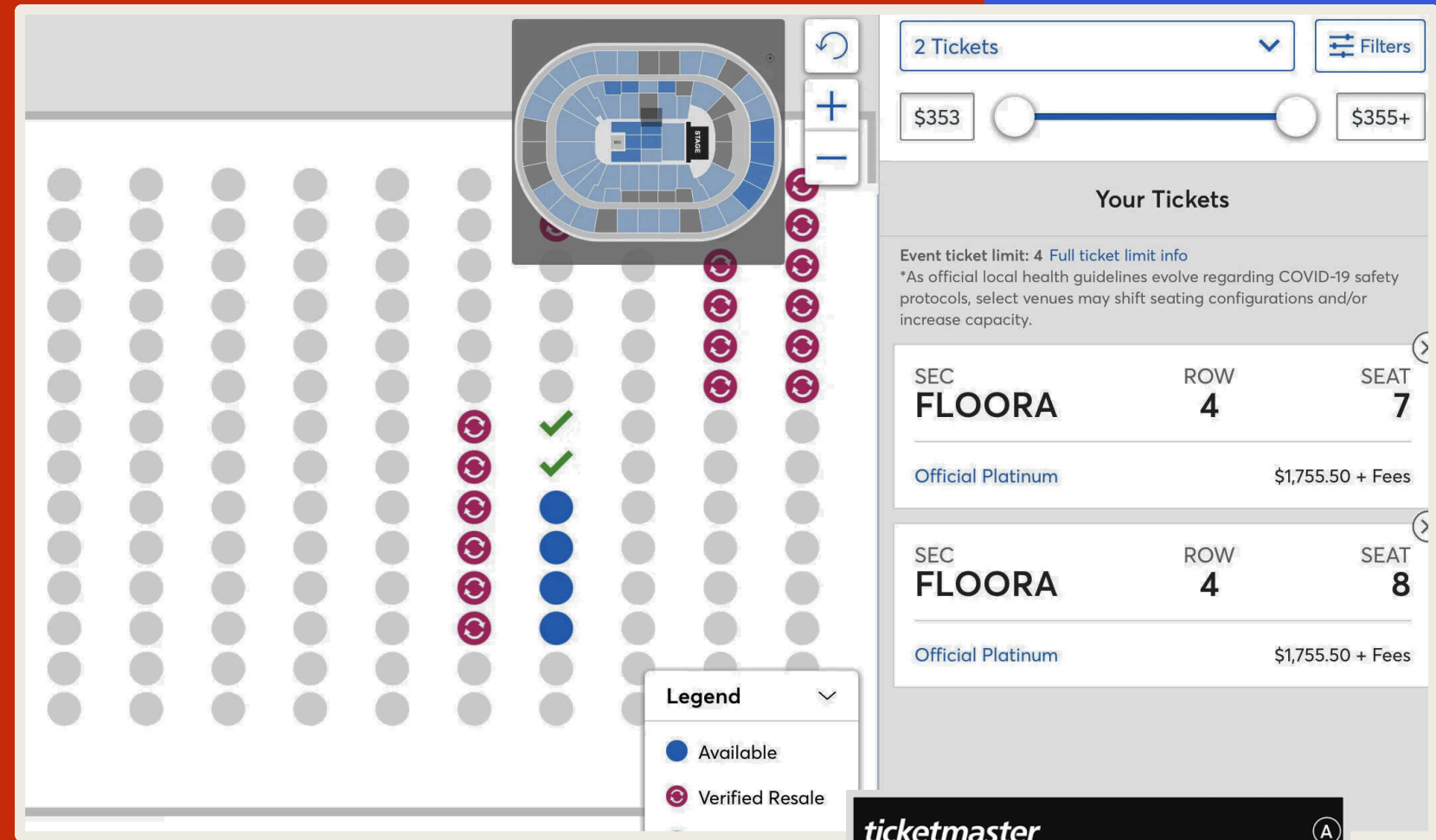
FINAL THOUGHTS

AMERICA HAS A (DYNAMIC) PROBLEM...

- Concerts, especially In the U.S. are rapidly becoming more and more expensive each year
 - Possible demand due to perceived “FOMO” post-pandemic
 - Rising production and insurance costs,
 - Lack of adequate regulation on ticketing industry → **ticket prices, both at face value and resale, increase.**

- **Major contributor to rise in cost is Ticketmaster’s “Dynamic Pricing” system.**

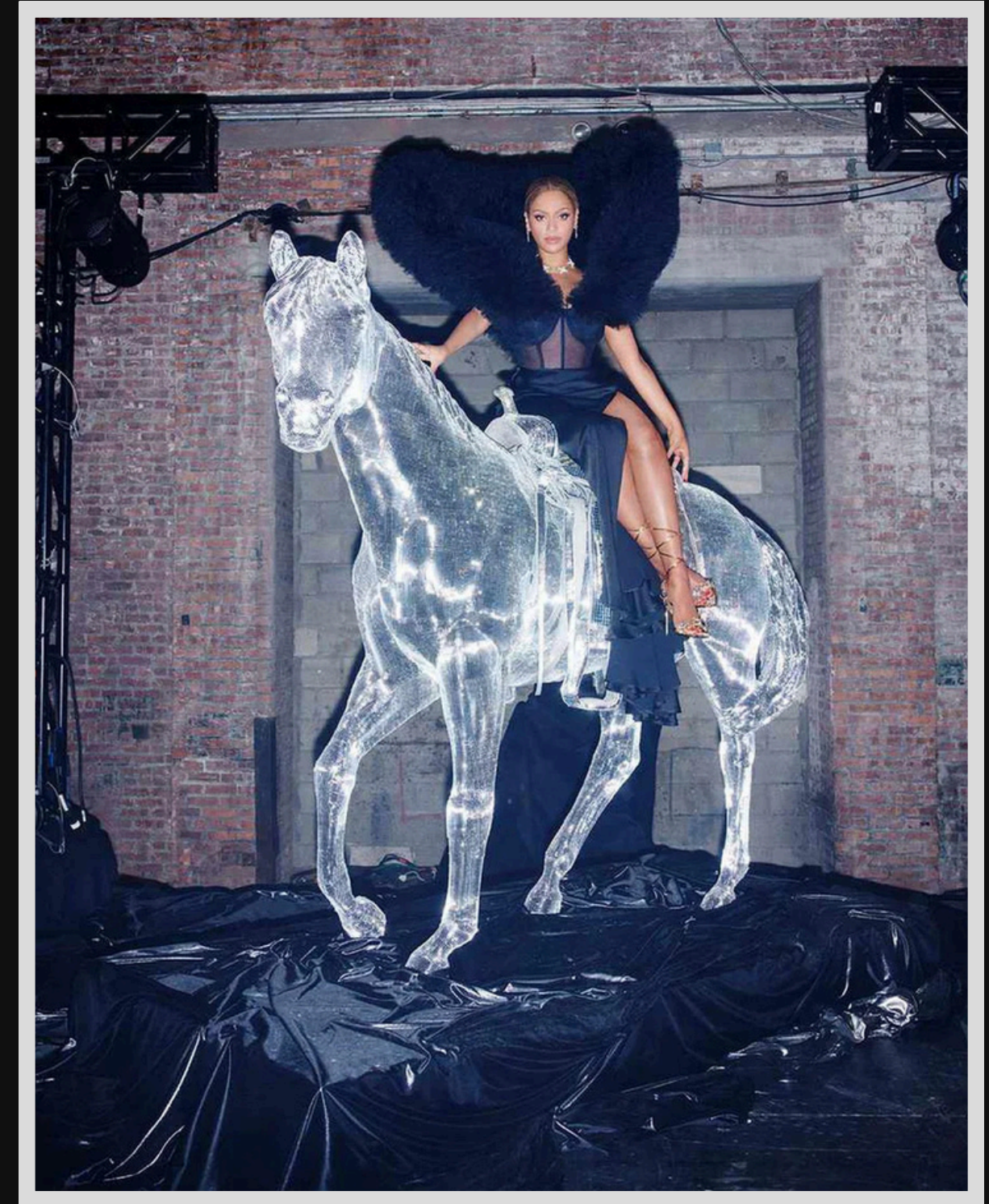
- The rollout has led to a wave of disgruntled fans, with accusations of price gouging, reliance on an abhorrent resale market, lack of transparency, and undermining the traditional sense of fairness and accessibility at live events.



WHY BEYONCÉ?

[...BECAUSE IT'S BEYONCÉ, DUH]

- Represents the intersection of **superstar status, high consumer expectations, and modern ticketing practices.**
- In response to the Eras Tour ticketing controversy, Ticketmaster introduced stricter verification and a staggered, lottery-based Verified Fan system for Beyoncé's Renaissance World Tour; **it would be interesting to also see whether these reforms had any impact on the secondhand market during the duration of Renaissance Tour.**
- With all things considered, **is there an optimal time to buy Beyoncé tickets for future shows in order to minimize the price paid?** This case study attempts to answer this question using detailed multi-city data from her 2023 tour.



METHODS & DATA

KEY VARIABLES:

- **EventDate:** The specific show date.
- **City:** The specific city.
- **Zone:** The broader seating area (ex. Upper Terrace).
- **Section:** Specific seating area within a zone (ex. 225C).
- **ActualPrice:** The ticket's asking price in USD.
- **PriceDate:** When the ticket was listed or the price was recorded.
- **Other Features:** Quantity available (Qty), Row, and any special zone designations (ex. "Club Renaissance").

The data compiled for this study was sourced from [SeatData.io](https://seatdata.io), an archive site that tracks third-party resale and primary listings.

For Beyoncé's Renaissance World Tour, only third party resale data was available.

The dataset covers dozens of her tour dates across North America, spanning various venues and cities.



METHODS & DATA

DERIVED VARIABLES:

- **DaysBeforeEventDate:** Calculated as the number of days between the listing date and the event date.
- *** Standardized Section Zones/Groups:** Mapped detailed section names into broader categories for comparison across venues.
- **Seat Quality Score:** Assigned based on the typical (median) price for each section group (process explained on next slide).

***Filtered out incomplete or ambiguous listings.**

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APPROACH

City	EventDate	PriceDate	PriceTime	DaysBeforeEventDate	Zone	Section	R...	Qty	StandardZone	Section_Group	Section_Num	ActualPrice	Lowest_Price	
<chr>	<chr>	<chr>	<chr>	<int>	<chr>	<chr>	<chr>	<int>	<chr>	<chr>	<int>	<dbl>	<dbl>	
Kansas City	2023-10-01	2023-09-27	4:52 PM		4	Upper Terrace	615	19	2	Upper Bowl	600s	615	413.1	63.00
Kansas City	2023-10-01	2023-09-27	12:31 PM		4	Club Renaissance	Club Renaissance		1	VIP Floor	Club Renaissance	NA	2700.0	1138.86
Kansas City	2023-10-01	2023-09-27	12:31 PM		4	Upper Terrace	617	31	3	Upper Bowl	600s	617	540.0	63.00
Kansas City	2023-10-01	2023-09-27	9:00 AM		4	Upper Terrace	615	19	2	Upper Bowl	600s	615	413.1	63.00
Kansas City	2023-10-01	2023-09-27	12:39 AM		4	Field	E	19	1	Floor	Floor	NA	1620.0	505.44
Kansas City	2023-10-01	2023-09-27	12:39 AM		4	Field	E	19	1	Floor	Floor	NA	1080.0	505.44
Kansas City	2023-10-01	2023-09-26	7:39 PM		5	Field	N	21	1	Floor	Floor	NA	1440.0	505.44
Kansas City	2023-10-01	2023-09-26	2:59 PM		5	Field	F	14	1	Floor	Floor	NA	855.0	505.44
Kansas City	2023-10-01	2023-09-26	2:59 PM		5	Upper Loge	310	11	2	Upper Bowl	300s	310	899.1	189.00
Kansas City	2023-10-01	2023-09-26	11:00 AM		5	Upper Terrace	639	29	1	Upper Bowl	600s	639	387.0	63.00

Section	R...	Qty	StandardZone	Section_Group	Section_Num	ActualPrice	Lowest_Price	ExpectedPrice_Reg	Deal_Score_Reg	Seat_Quantity_Score	Final_Deal_Score
<chr>	<chr>	<int>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
615	19	2	Upper Bowl	600s	615	413.1	63.00	178.03	0	0.3	0
Club Renaissance		1	VIP Floor	Club Renaissance	NA	2700.0	1138.86	1415.70	0	0.8	0
617	31	3	Upper Bowl	600s	617	540.0	63.00	178.03	0	0.3	0
615	19	2	Upper Bowl	600s	615	413.1	63.00	178.03	0	0.3	0
E	19	1	Floor	Floor	NA	1620.0	505.44	833.96	0	0.7	0
E	19	1	Floor	Floor	NA	1080.0	505.44	833.96	0	0.7	0
N	21	1	Floor	Floor	NA	1440.0	505.44	833.96	0	0.7	0
F	14	1	Floor	Floor	NA	855.0	505.44	833.96	0	0.7	0
310	11	2	Upper Bowl	300s	310	899.1	189.00	273.12	0	0.2	0
639	29	1	Upper Bowl	600s	639	387.0	63.00	178.03	0	0.3	0

METHODS & DATA

The data compiled for this study was sourced from [SeatData.io](#), an archive site that tracks third-party resale and primary listings.

For Beyoncé’s Renaissance World Tour, only third party resale data was available.

The dataset covers dozens of her tour dates across North America, spanning various venues and cities.

- # New columns for df - csv file
- df <- df %>%
 - mutate(
 - StandardZone = mapply(standard_zone, Zone, Section),
 - Section_Group = sapply(Section, section_group),
 - Section_Group = ifelse(StandardZone == "Floor", "Floor", Section_Group),
 - Section_Num = extract_section_num(Section),
 - PriceDate = as.Date(PriceDate, format = "%m-%d-%y"),
 - EventDate = as.Date(EventDate, format = "%Y-%m-%d"),
 - DaysBeforeEventDate = as.numeric(EventDate - PriceDate)
 -) %>%
 - filter(Section_Group != "Other") %>%
 - mutate(row_id = row_number()) %>%
 - rename(ActualPrice = Price)

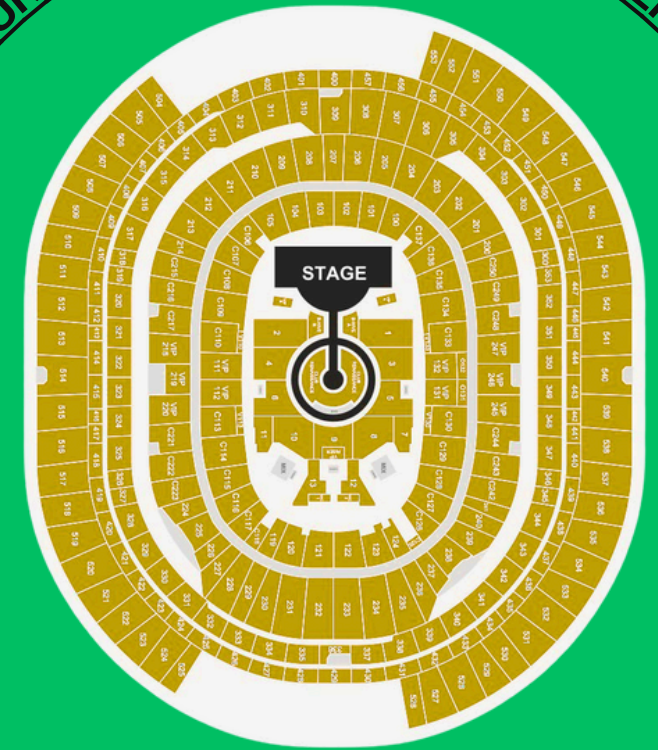
CODE+DF
SNIPPET:



*Filtered out incomplete or ambiguous listings.

Section Group	Location	Experience	Price
VIP Pure Honey Risers	Raised VIP platforms on the main stage	Exclusive fan interactions with artist Best views and perks with seats provided	\$\$\$\$\$\$
B-Hive	VIP Pit located at very front of the main stage (two sections; one on each side of the main stage)	Intimate, special fan zone for the enthusiasts Standing-room only, with a private bar and/or amenities	\$\$\$\$\$
Club Renaissance	VIP Pit located between main stage	Fans who want to be as close to the artist as possible, with a special “club-like” party experience Standing-room only, with a private bar and/or amenities	\$\$\$\$
Floor	Main flat area in front of the stage (not VIP)	Still close enough, but without VIP perks Seats provided (US Only)	\$\$\$\$
Lower Bowl (“100s”)	1st tier of permanent seats surrounding the floor	Good view, comfortable seat Less likely to have obstructed view compared to floor due to raked seating arrangements	\$\$\$
Club (“200s”)	2nd tier, sometimes with extra amenities (buffet, premium quality seats, etc.)	Slightly further, though can be considered more comfortable.	\$\$
*Upper Bowl (“300s+”) *Some venues have sections that go up to 700	Highest, farthest seats in the arena; budget-friendly	Widest view. Great for seeing entire production or just to enjoy the music/atmosphere	\$

MEASURING SEAT COMFORT AND EXPERIENCE



UPPER BOWL

COZY IN THE CROWD*

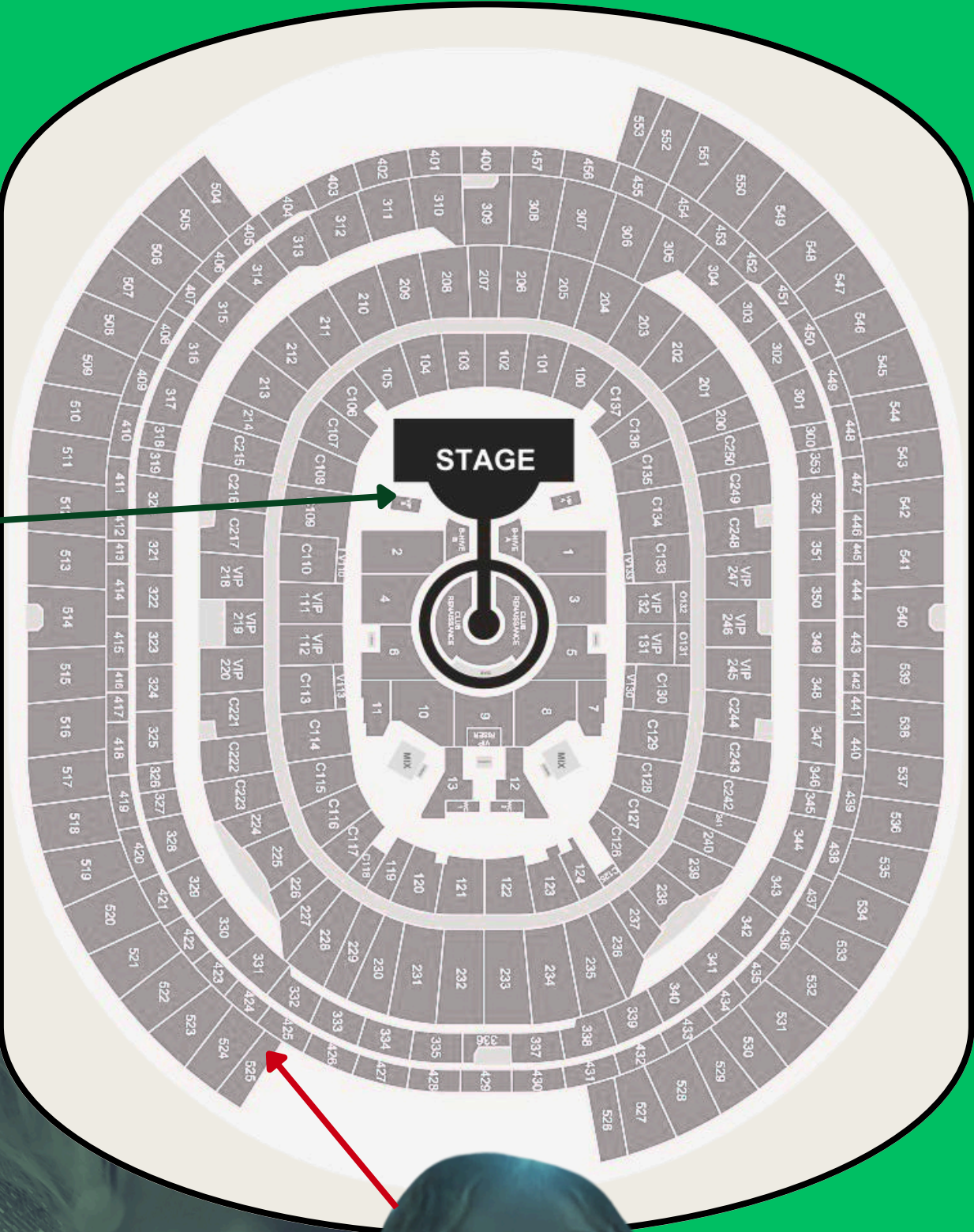
(Sections Explained)

***WHY THIS APPROACH?**

ADJUSTS FOR VENUE DIFFERENCES: PRICES ARE COMPARED FAIRLY ACROSS CITIES.

DATA-DRIVEN: BASED ON REAL MARKET DATA, NOT ASSUMPTIONS. (THOUGH STILL NOT 100% OBJECTIVE)

FAIR DEAL SCORES: QUALITY SCORE ENSURES DEAL RATINGS REFLECT BOTH PRICE AND SEAT DESIRABILITY



STEP-BY-STEP SUMMARY

SECTION STANDARDIZATION:

SEATS WERE GROUPED INTO STANDARDIZED SECTIONS (E.G., FLOOR, LOWER BOWL, CLUB, ETC.) USING SECTION LABELS AND CODES.

EXPECTED PRICE MODELING:

A RIDGE REGRESSION MODEL (WITH LOG-TRANSFORMED PRICES) PREDICTED TYPICAL TICKET PRICES FOR EACH SECTION AND CITY, ACCOUNTING FOR LOCAL PRICING DIFFERENCES.

SECTION MEDIAN PRICE:

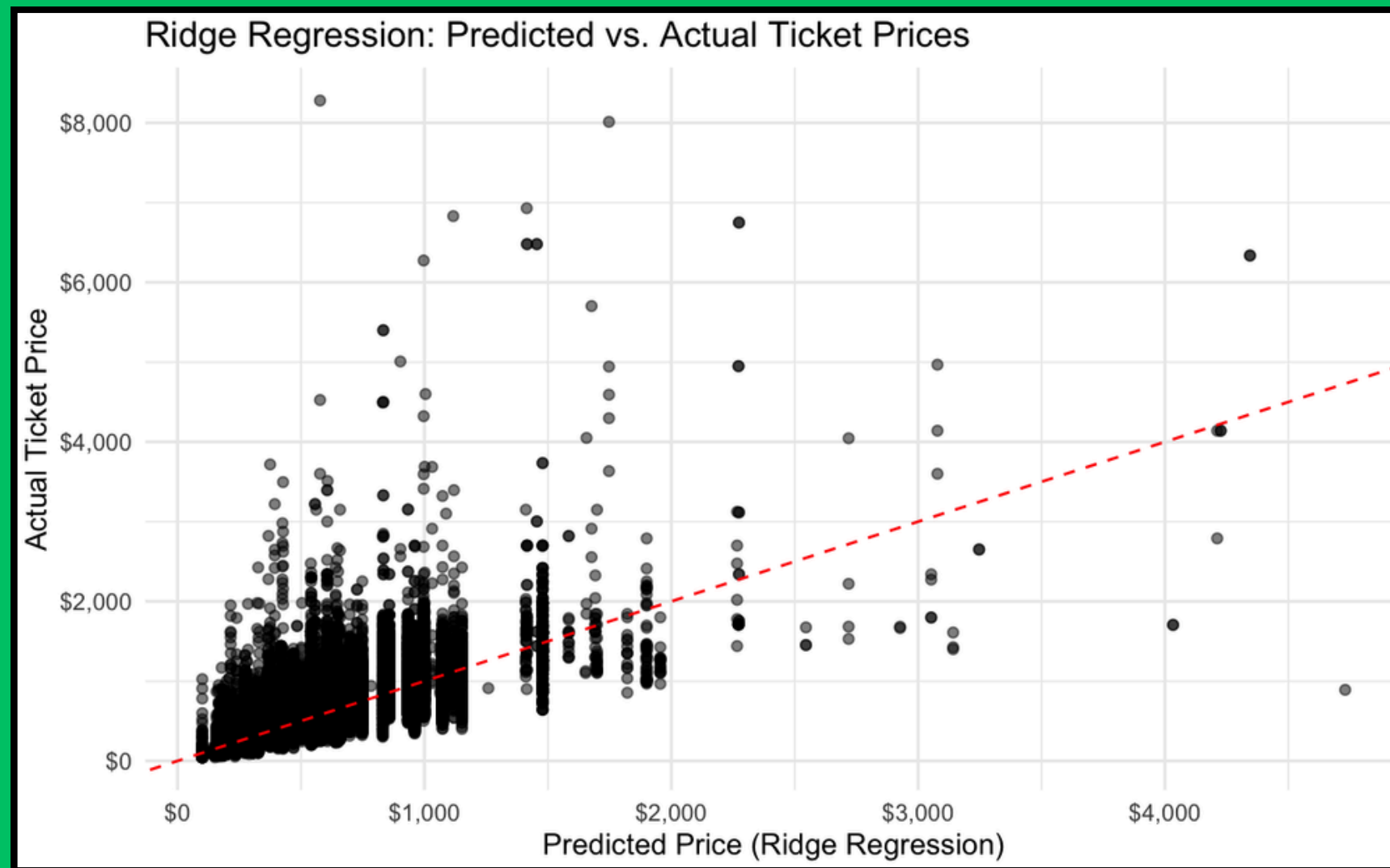
FOR EACH SECTION GROUP, WE CALCULATED THE MEDIAN PREDICTED PRICE TO REPRESENT THE “USUAL” COST.

SEAT QUALITY SCORE:

SECTIONS WERE RANKED BY THEIR MEDIAN PRICE. THE SEAT QUALITY SCORE IS EACH SECTION’S PERCENTILE RANK (FROM 0.0-1.0).

COZY IN THE CROWD*

HIGHER SCORES = BETTER, MORE DESIRABLE SECTIONS
LOWER SCORES = WORSE, LESS APPEALING SECTIONS



Most points near the dashed line: Model predictions generally match actual prices for typical tickets.

Wide vertical spread at high prices: Model underestimates rare, very expensive tickets (outliers).

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COZY IN THE
CROWD*
- RRM

DIVA DISCOUNTS:

DEAL SCORES: WHAT THEY MEASURE AND
HOW THEY ARE CALCULATED

Deal_Score_Reg:

- How much of a bargain was this ticket?
 - Compares price paid (ActualPrice) to the typical price for that section, city, and event (ExpectedPrice).
 - High score = paid much less than average (big bargain).
 - $Deal_Score_Reg = (ExpectedPrice_reg - ActualPrice) / ExpectedPrice_reg$

Seat_Quality_Score:

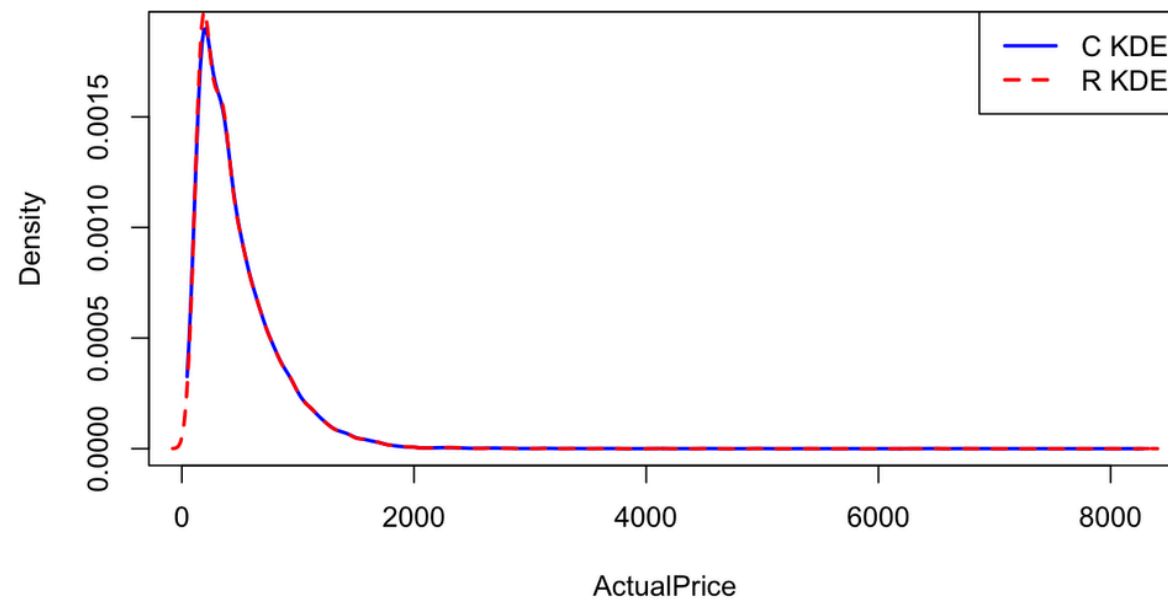
- How good is this section? - see previous slide(s) for greater detail/reference

Final_Deal_Score:

- How great is this ticket overall?
 - Combines both the bargain and seat quality using the geometric mean.
 - High score = ticket is both cheap and in a great section.
 - $Final_Deal_Score = \sqrt{Deal_Score_Reg * Seat_Quality_Score}$



Univariate KDE: C (blue) vs R (red)



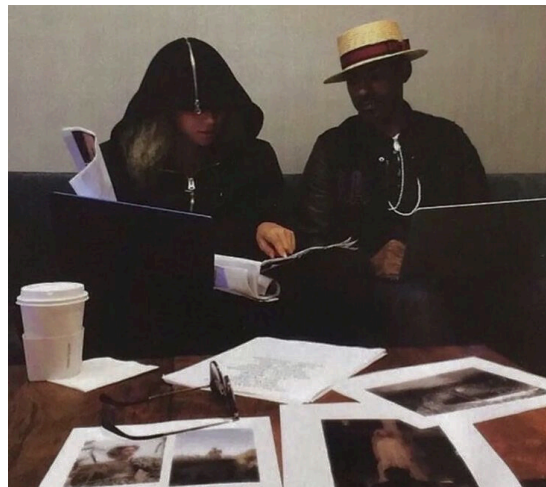
Most tickets sold at low prices; right-skewed distribution.

Few tickets sold at very high prices (outliers).

- Prices not normally distributed; heavy upper tail.
- Log transformation QQ reduces skewness but some outliers remain.

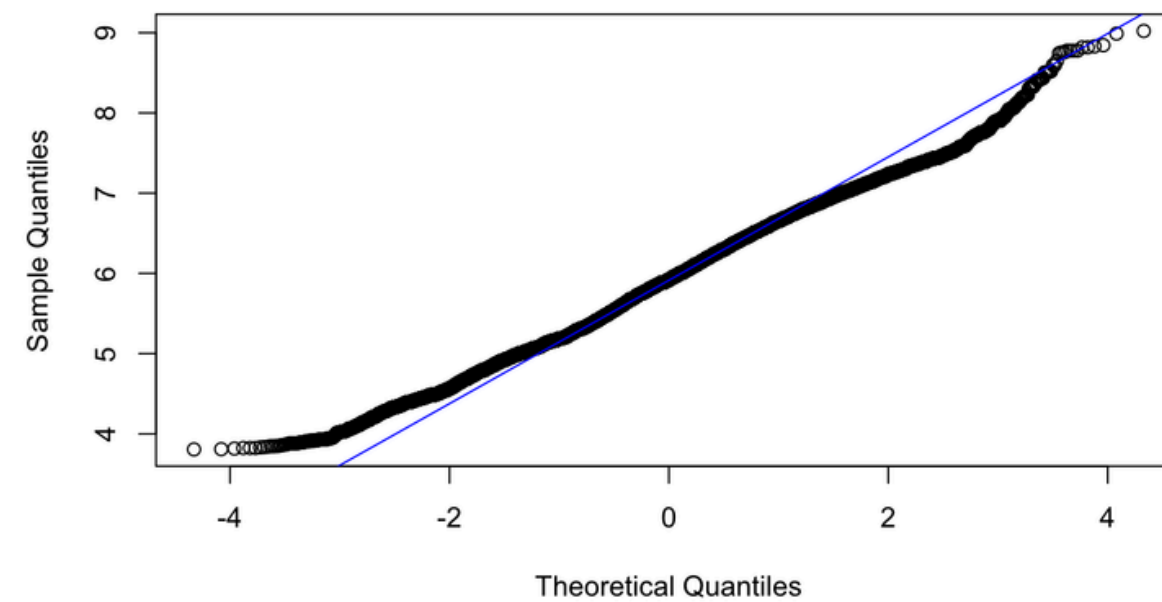
Key Visuals

KDE & QQ

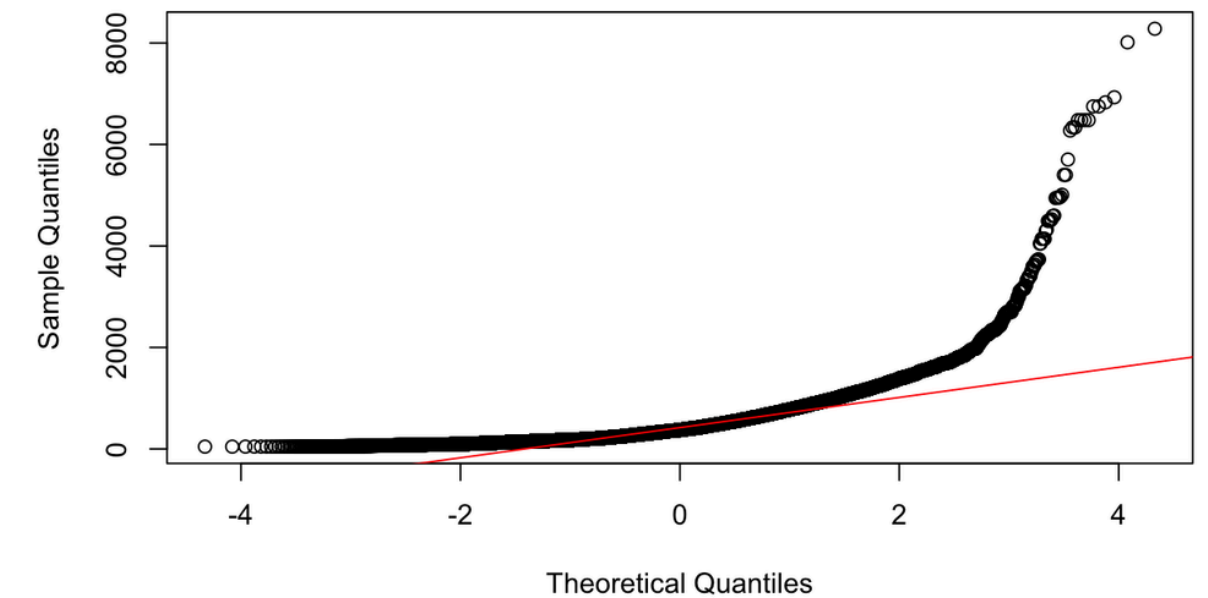


Results

QQ Plot of Log(Ticket Prices)



QQ Plot of Ticket Prices



Summary Stats



- Min. 1st Qu. Median Mean 3rd Qu. Max.
- 45.0 220.5 373.1 473.2 622.1 8280.0
- 1% 5% 25% 50% 75% 95% 99%
- 82.5600 124.5300 220.5000 373.0900 622.0825 1135.2800 1619.0949
- [1] 473.2497
- [1] 359.4963
- [1] 401.5825
- [1] 45 8280

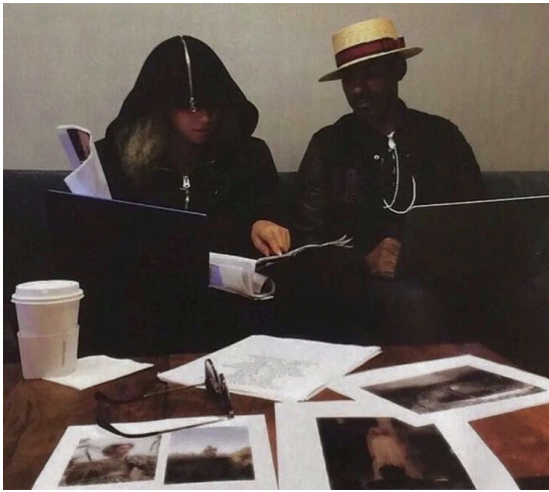
Section group is the main driver of ticket price and deal quality.

- Standard sections (100s–700s):
 - Prices are more consistent, but some outliers (>\$2,000) exist.
 - Occasionally, strong deals are found, but they are rare.
- Premium/VIP sections (Floor, Club Renaissance, B-Hive):
 - Prices are much higher and more variable.
 - These sections show both the highest prices and the greatest spread in deal quality.
- Some exceptional bargains (“red dot” outliers) occur even in premium sections.

Every section has rare outliers: very high prices (overpays) or exceptional deals (underpays).

Key Visuals

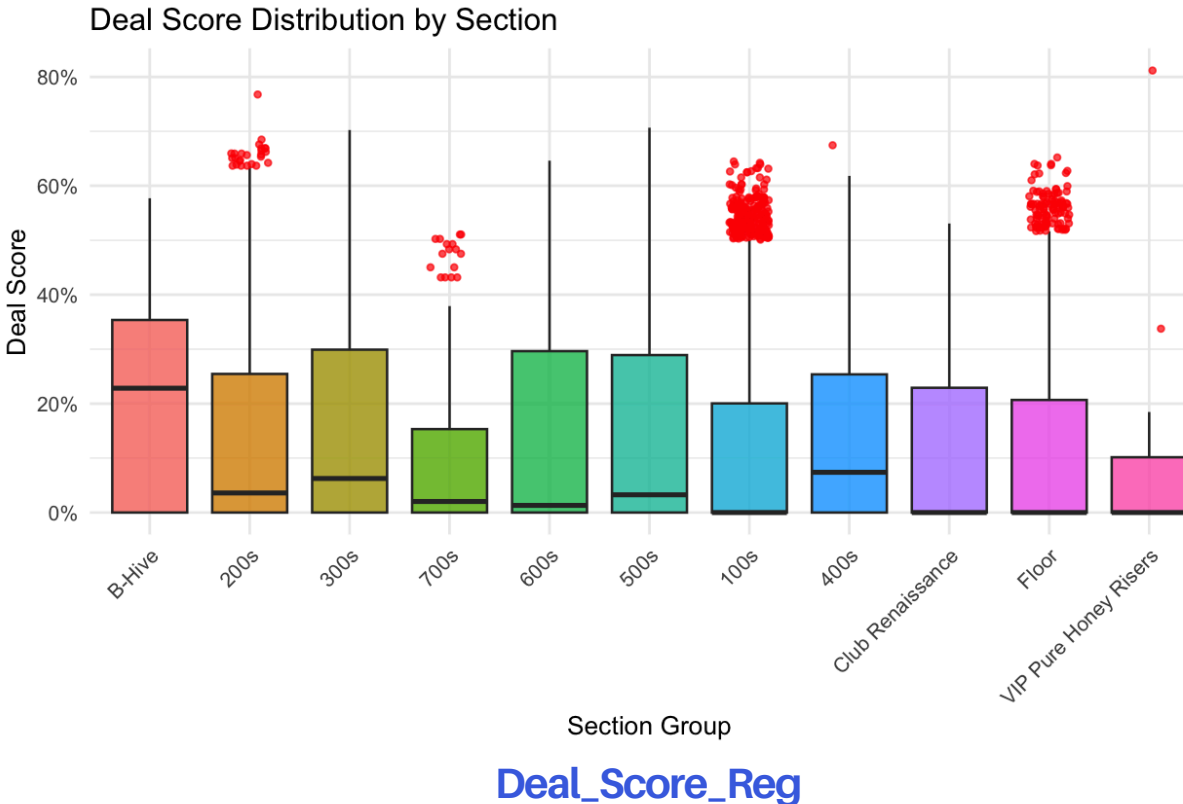
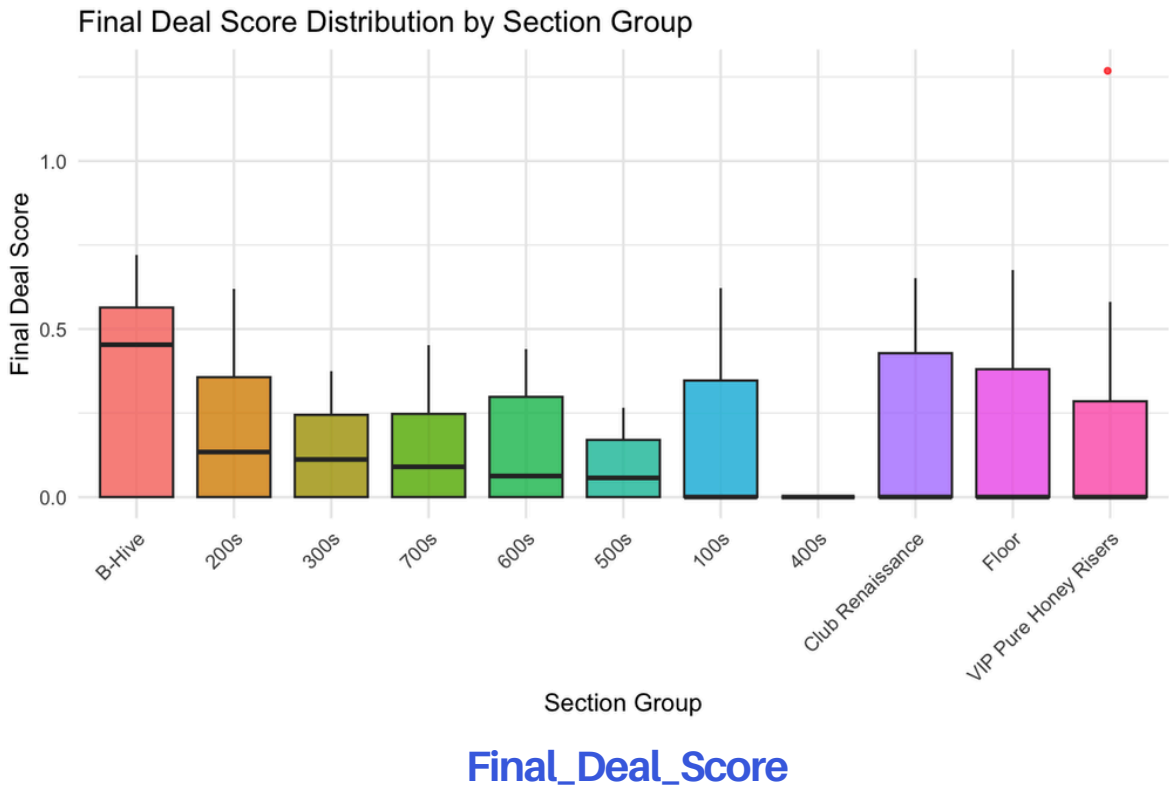
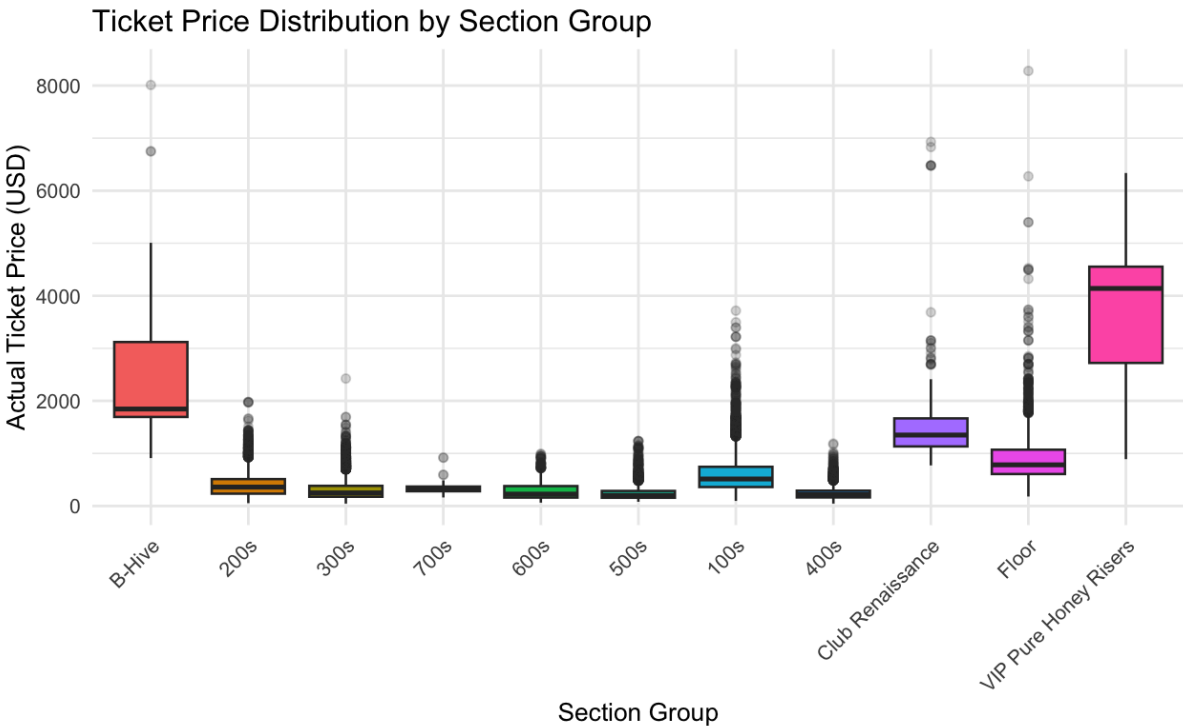
BOXPLOTS



Summary Stats



Results



Ticket Price & Deal Score Distribution by Section

- Standard seating sections (100s–700s): Median \$200–\$513
- Floor: Median \$782
- Club Renaissance: Median \$1,350
- B-Hive: Median \$1,845
- Most buyers paid under \$1,000 except in premium sections
- Outliers (very high prices) exist in every section

Price Changes as Event Approaches

Best chance (55-60%) for a top deal: Right after tickets go on sale (200+ days out).

- VIP/Premium sections: Prices start very high, dip 50-100 days before the event, then rise again close to showtime (U-shaped pattern).
- Standard sections (100s-700s): Prices are much lower and remain flat over time, with no big last-minute changes.

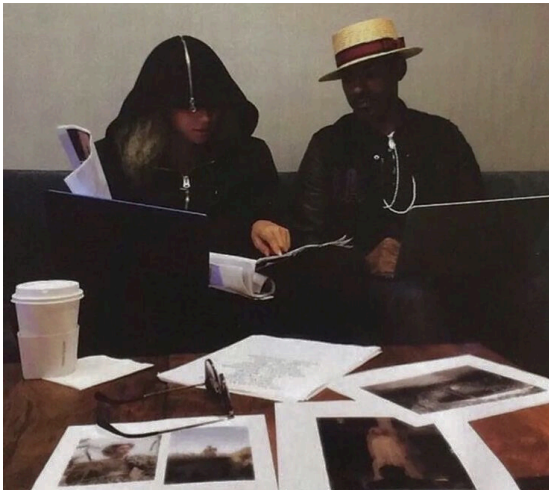
Overall, ~20% dip soon after onsale
For premium seats, the best deals are usually in the middle of the sales window.

For regular sections, timing matters less; prices are steady.

But overall, buy early for the best odds of a great deal esp. if Premium tickets are not a need. Waiting until the last minute offers only a small chance of improvement.

Key Visuals

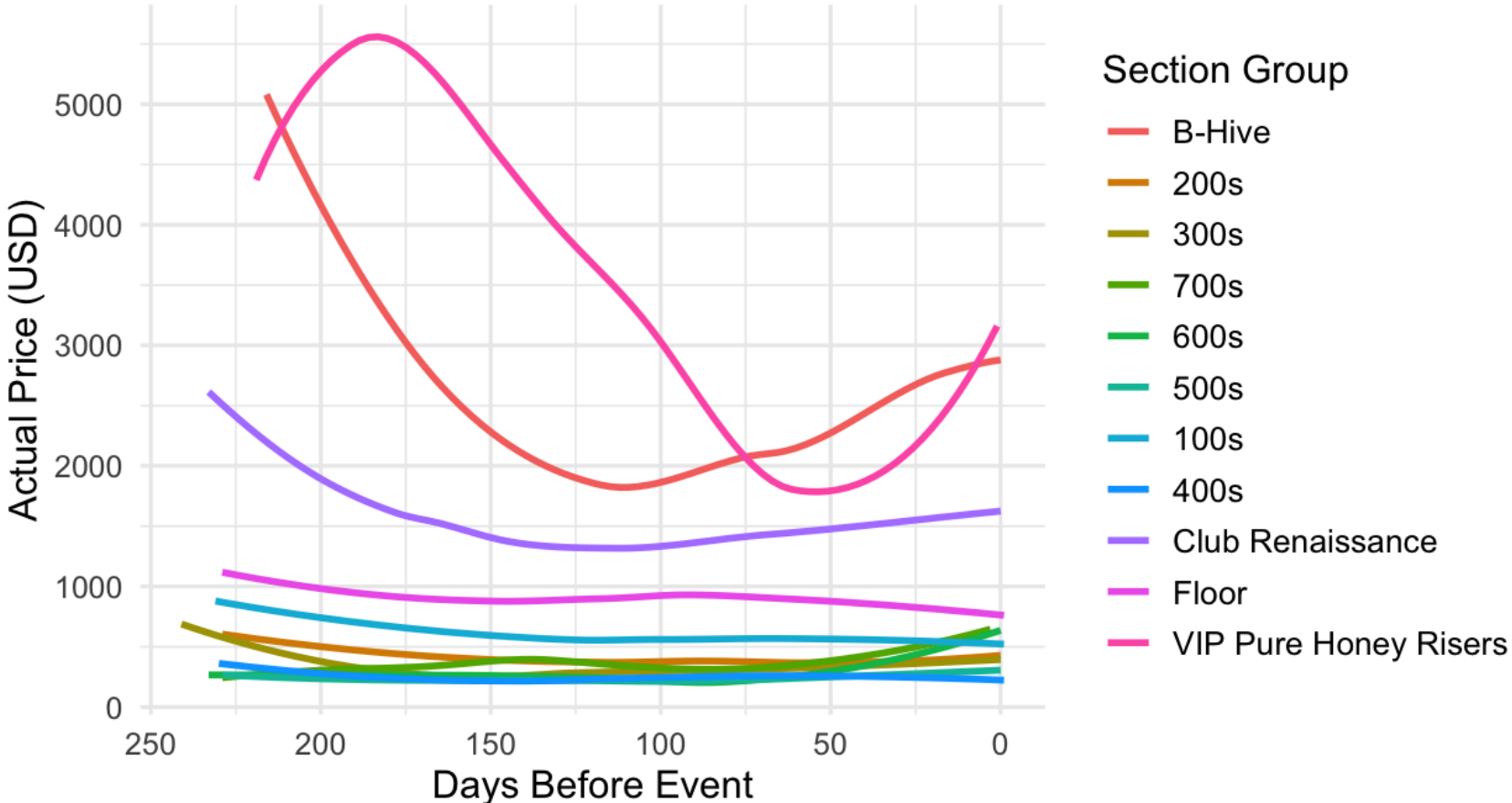
LOESS
/GAM



Results

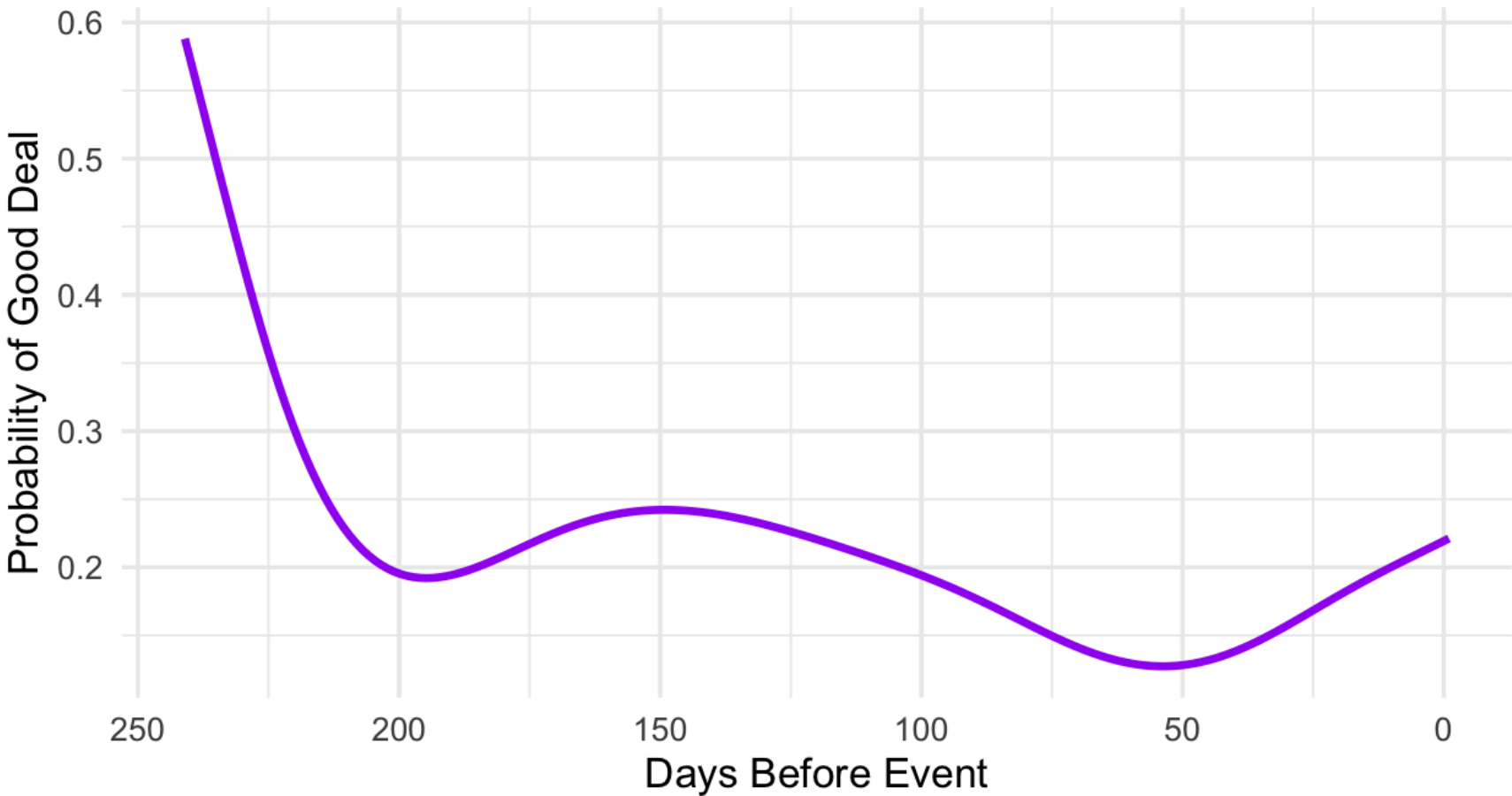
LOESS-Smoothed Ticket Price Trends by Section Group

Aggregated across all cities



Probability of Getting a Good Deal vs. Days Before Event

GAM (logistic kernel regression); good deal = top 20% of all-in scores



LIMITATIONS

MODELS, CODE, AND THRESHOLDS DEVELOPED HERE MAY REQUIRE REVISION FOR FUTURE EVENTS, AS TICKETING MARKETS, DATA SOURCES, AND BUYER BEHAVIOR EVOLVE.

CHANGES IN ARTIST DEMAND, EVENT STRATEGIES, OR EXTERNAL FACTORS COULD ALSO SIGNIFICANTLY ALTER TICKET PRICING PATTERNS AND DEAL OPPORTUNITIES.



Real-world events (artist announcements, tour changes, local news) can have big impacts on prices, which aren't captured here.

Example: Cowboy Carter World Tour — Market Saturation or Scarcity
If an artist performs fewer shows in a region in the future, ticket supply and demand, and thus prices, will change in ways my current analysis cannot anticipate.

CONCLUSION



- Based on the data and ridge regression model, **ticket prices for Beyoncé's tour are generally lowest between 2 to 4 weeks before the event, particularly a high quality one.**
 - During this window, there's a higher concentration of "good deal" tickets compared to **very early (right after onsale) or last-minute purchases.**
 - **Buy early for most in-demand/premium sections and cities**
- **Prices tend to be higher just after tickets are released and can spike again in the final days leading up to the concert, especially for high-demand sections.**

There is no perfect time,
just the best time.
Stalking Ticketmaster all
day never hurts.

THANK
YOU

PRESENTED BY
MERCY EME